PROGRAM 1: Generate Text Embeddings and Perform Semantic Similarity Search

AIM

To generate text embeddings using a Sentence Transformer model and find semantically similar sentences using cosine similarity.

ALGORITHM

1. Install the required libraries.
2. Import the Sentence Transformer model and cosine similarity function.
3. Load a pre-trained sentence embedding model.
4. Convert the input sentences into embeddings.
5. Convert the query into an embedding.
6. Calculate cosine similarity between the query and sentences.
7. Display the similarity scores and most similar sentence.

CODE

Cell 1: Install the library

```
!pip -q install sentence-transformers
```

Cell 2: Python program

```
from sentence_transformers import SentenceTransformer
from sklearn.metrics.pairwise import cosine_similarity
# Load the model
model = SentenceTransformer("all-MiniLM-L6-v2")
# Input sentences
sentences = [
    "Artificial intelligence helps machines learn.",
    "Python is a programming language.",
    "Machine learning is a branch of artificial intelligence.",
```

```
    "The weather is sunny today."
]
# Convert sentences into embeddings
embeddings = model.encode(sentences)
# Input query
query = "What is machine learning?"
# Convert query into embedding
query_embedding = model.encode([query])
# Calculate cosine similarity
scores = cosine_similarity(query_embedding, embeddings)[0]
# Display results
print("Query:", query)
print("\nSemantic Similarity Results:")
for i, score in enumerate(scores):
    print(f"{sentences[i]} -> {score:.4f}")
best_index = scores.argmax()
print("\nMost Similar Sentence:")
print(sentences[best_index])
```

INPUT

Query: What is machine learning?

OUTPUT

Illustrative output; the exact similarity scores may vary.

Query: What is machine learning?

Semantic Similarity Results:

Artificial intelligence helps machines learn. -> similarity score

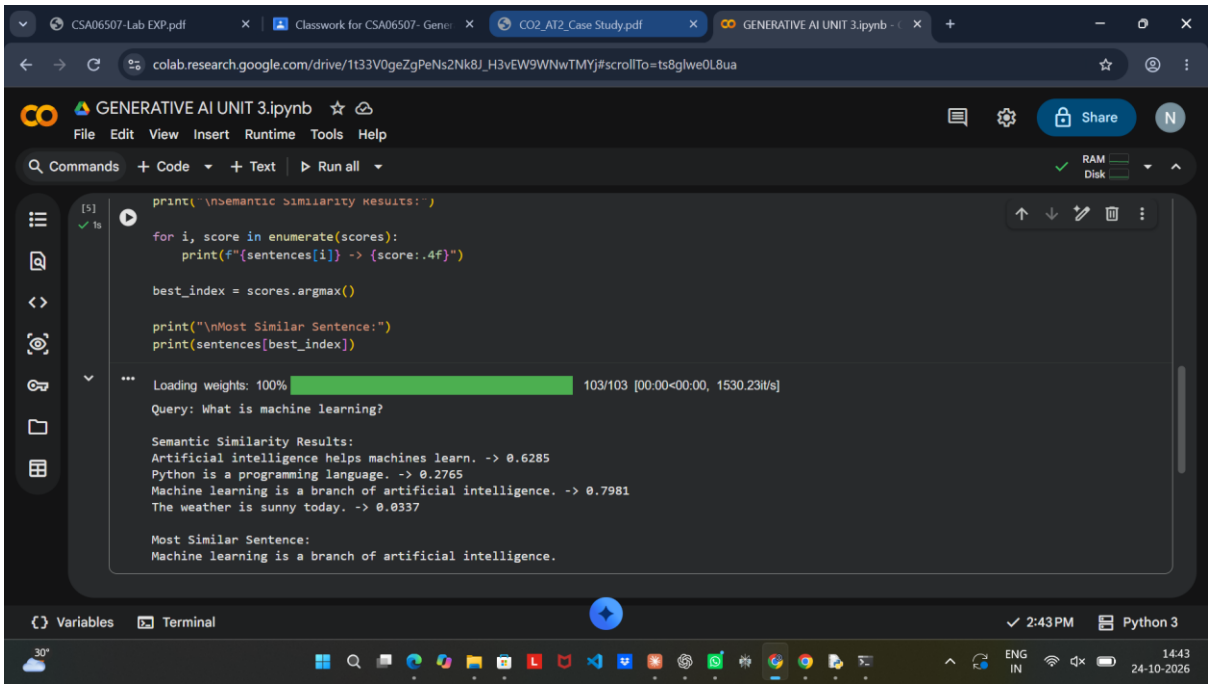
Python is a programming language. -> similarity score

Machine learning is a branch of artificial intelligence. -> similarity score

The weather is sunny today. -> similarity score

Most Similar Sentence:

Machine learning is a branch of artificial intelligence.



RESULT

Text embeddings were generated successfully, and the most semantically similar sentence was identified using cosine similarity.